Adding and subtracting fractions

1 Simplify:

a $\frac{1}{11}+\frac{9}{11}$ b $\frac{2}{6}+\frac{2}{6}$ c $\frac{4}{7}-\frac{2}{7}$ d $\frac{5}{6}-\frac{3}{6}$

e $\frac{7}{9}+\frac{4}{9}$ f $\frac{8}{9}-\frac{7}{9}$ g $\frac{7}{25}+\frac{3}{25}$ h $\frac{9}{22}-\frac{5}{22}$

2 Simplify the following, writing your answer as a mixed numeral where necessary:

a $\frac{1}{4}+\frac{1}{9}$ b $\frac{6}{7}+\frac{1}{14}$ c $\frac{1}{4}+\frac{1}{6}$ d $\frac{1}{5}+\frac{2}{4}$

e $\frac{7}{8}+\frac{5}{6}$ f $\frac{6}{5}+\frac{5}{3}$ g $\frac{1}{3}-\frac{1}{10}$ h $\frac{3}{4}-\frac{1}{8}$

i $\frac{1}{9}-\frac{1}{12}$ j $\frac{2}{8}-\frac{1}{9}$ k $\frac{4}{7}+\frac{1}{3}$ l $\frac{2}{9}-\frac{1}{18}$

3 Simplify the following, writing your answer as a mixed numeral where necessary:

a $\frac{8}{5}+\frac{11}{5}$ b $\frac{7}{6}+\frac{13}{10}$ c $\frac{5}{2}+3$ d $2\frac{3}{11}+4\frac{7}{11}$

e $3\frac{2}{6}+4\frac{4}{5}$ f $\frac{46}{22}-\frac{16}{22}$ g $\frac{9}{4}-\frac{6}{14}$ h $\frac{7}{4}-\frac{5}{3}$

i $\frac{10}{8}-\frac{7}{6}$ j $4-\frac{7}{9}$ k $5\frac{10}{11}-4\frac{8}{11}$ l $4\frac{6}{7}-3\frac{6}{8}$

m $3\frac{1}{3}+4\frac{11}{12}$ n $6\frac{1}{2}-4\frac{1}{10}$ o $6\frac{1}{2}-\frac{3}{8}$ p $2\frac{2}{3}+3\frac{11}{15}$

4 Simplify the following, writing your answer as a mixed numeral where necessary:

a $1+\frac{1}{2}$ b $2-\frac{3}{4}$ c $6+1\frac{4}{5}$ d $8-2\frac{1}{2}$

e $7+3\frac{3}{4}$ f $10-5\frac{1}{3}$ g $9-2\frac{5}{7}$ h $11+4\frac{3}{8}$

5 Simplify the following, writing your answer as a mixed numeral where necessary:

a $\frac{3}{4}+\frac{1}{8}-\frac{1}{4}$ b $\frac{5}{9}+\frac{2}{3}+\frac{5}{6}$ c $\frac{10}{12}-\frac{1}{6}-\frac{1}{4}$ d $\frac{8}{10}+\frac{2}{5}-\frac{7}{20}$

e $\frac{2}{5}+\frac{1}{10}-\frac{7}{30}$ f $\frac{7}{45}+\frac{2}{9}-\frac{8}{9}$ g $\frac{4}{45}+\frac{7}{5}-\frac{8}{5}$ h $\frac{3}{4}+\frac{7}{8}-\frac{5}{24}$

i $\frac{6}{35}-\frac{1}{7}+\frac{3}{14}$ j $\frac{2}{45}+\frac{7}{9}+\frac{2}{3}$ k $\frac{1}{2}-\frac{1}{3}-\frac{1}{9}$ l $\frac{8}{21}-\frac{2}{7}+\frac{9}{14}$

Applications

6 Marge ran three ninths of a kilometre. She took a 5-minute rest, then ran another three thirds of a kilometre. How many kilometres did she run in total?

7 Jack felt dehydrated. He drank one third of a litre at first, then drank another one ninth of a litre. How many litres did he drink in total?

8 Valerie walks eight sixths of a kilometre to school while Fred walks one half of a kilometre to school. How much farther does Valerie walk than Fred?

9 A bridge is being built to span $\frac{3}{4}$ km. After a month, the engineer reported that $\frac{3}{8}$ km has been completed so far. What part of the bridge has not yet been completed?

10 Jane eats $\frac{1}{4}$ of a cake and Joshua eats $\frac{1}{3}$ of the cake. What fraction of the cake did they eat altogether?

11 During a blizzard it snowed $10\frac{1}{2}$ cm. When the sun came out the next day, $5\frac{3}{4}$ cm of show melted. How much snow was left on the ground?

12 A market stall is open for $3\frac{1}{2}$ hours in the morning and $2\frac{3}{4}$ hours in the evening. How many hours is the stall open altogether?

13 Georgio buys 3 cups of sugar to use in his baking. He uses $\frac{3}{4}$ of a cup in his cake, $1\frac{1}{5}$ of a cup in his pastries and accidentally spilled $\frac{1}{2}$ of a cup. How much sugar does he have left?

14 Mr. Pemberton, the president of Zoom Motors, owns $\frac{1}{3}$ of the stock of the company. His wife owns $\frac{1}{4}$ of the stock. What fractional part of the stock will his daughter need to purchase in order for the family to own $\frac{3}{4}$ of the stock?

15 Missy mailed 4 letters that have a total weight of 3 grams. One letter weighed $\frac{1}{2}$ of a gram, another weighed $\frac{3}{4}$ of a gram, and a third weighed $\frac{7}{8}$ of a gram. How much did the fourth letter weigh?